Civis Terra: Construction Material Quantity Estimation and digital tracking for Waste Reduction

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Team Name: *Civis Terra*

Domain: *Space , Aerospace , Mobility and Next generation infrastructure*

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Presentation Outline

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Problem Statement

Construction sites may order or use more materials than needed. Wrong measurements, design changes, leftover mortar and poor tracking can waste cement, sand, bricks and concrete. This increases costs and sends usable materials to waste.

Problem Understanding

- **Who faces it?** Site engineers, contractors, masons, workers and project owners.
- **Causes:** Estimation errors, design changes, over-ordering, poor storage and extra mortar prepared to avoid delays.
- **Effects:** Higher costs, delays, extra transport and unnecessary use of natural resources.
- **Site insight:** Our preliminary illustrative mason conversation highlighted that quantities are estimated from drawings, but measurements can change and extra mortar may be prepared.

Existing Gaps & Challenges

- Material usage is often recorded manually, making it hard to update and compare.
- Initial estimates may not reflect changes during construction.
- Research explains many causes of waste, but site teams need a simple tool to track usage and receive early warnings.
- **Challenge:** Keep the tool easy to use and let users update quantities as work changes.

Our Proposed Solution

- **Civis Terra** is a proposed digital tool to estimate, track and help reduce construction material waste.
- Users enter measurements and planned quantities, then record materials received, used and left over.
- The app compares planned quantities with actual usage and flags possible overuse or excess stock.
- It suggests practical actions such as checking measurements, improving storage and reusing suitable leftovers.
- **Value:** Help site teams make better decisions, save money and reduce waste.

Technical Approach

Project measurements and details

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Quantity estimate → Stock and usage log

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Compare planned vs. actual use

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Waste warning, summary and suggestions

First version uses manual entries and simple calculations. Later versions could learn site-specific waste patterns from collected data.

Proposed Tech Stack

Frontend:	HTML, CSS, JavaScript
Backend:	Python with Flask or a lightweight API
Database:	Firebase or SQLite
AI/ML:	Future option to learn waste patterns from project records

Innovation & Uniqueness

- Combines quantity estimation with daily material-in and material-out tracking.
- Aims to flag possible waste while work is happening, not only after the project ends.
- Gives workers and site engineers practical suggestions they can act on.
- Can start with manual entries and become smarter as more project data is collected.

Feasibility & Scalability

- **Feasibility:** A prototype can be built as a mobile-friendly website with forms, quantity calculations and a small database.
- **Scalability:** Add more materials, multiple projects, reports and site-specific waste factors.
- **Risks:** Incorrect entries and changing site conditions may affect estimates.
- **Mitigation:** Let users update measurements, show assumptions and treat alerts as guidance rather than exact predictions.

Cost for Product (Prototype Estimate)

- **Development:** Student-built prototype using free tools: 0 software licence cost.
- **Hosting/database:** Free tiers or a local database can be used for initial testing.
- **Estimated prototype cost:** 0–2,000, assuming team laptops and free service tiers. This is an approximate estimate, not a quotation.
- **Possible future costs:** Domain, paid hosting, backups, support or optional sensors.

Use Cases (3 Examples)

- **Site engineer:** Enters wall dimensions and estimates bricks and mortar materials before ordering.
- **Storekeeper:** Records materials received and issued, helping compare stock with usage.
- **Mason or supervisor:** Records leftover mortar or unused materials and uses the record to improve planning for the next task.

Impacts and Benefits

- **Economic:** Reduce unnecessary purchases and improve control of material costs.
- **Environmental:** Reduce avoidable construction waste and demand for raw materials.
- **Practical:** Keep a clear record of materials ordered, used and left over.
- **Measurement:** During a pilot, compare planned quantities, actual usage and leftovers to see whether waste decreases.

Sustainable Development Goals (SDGs)

SDG	Target	Goal Name	Connection
12	12.5	Responsible Consumption and Production	Track material use and help prevent, reduce and reuse construction waste.
11	11.6	Sustainable Cities and Communities	Support better management of waste from urban construction.
9	9.4	Industry, Innovation and Infrastructure	Improve resource efficiency through a practical digital tool.

Individual Member Contribution

Member	Contribution
[S Neal Anto]	Research papers, citation review and causes of material waste.
[Boopeshwar S]	Site problem understanding and user needs.
[Sasikiran Reddy]	App features, interface and material-tracking workflow.
[Vishnu Prasad M V]	Technical approach, testing, presentation and documentation.

Replace names and tasks to match each member's actual contribution.

Research Basis

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Confirm bibliography details and citation counts before submission. The mason conversation should be labelled as illustrative unless it was actually conducted.

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Civis Terra

Smarter material planning. Less construction waste.